TEMPLATE GUIDE

OUC Smart Beamer Template

User Guide and Reference

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Template guide for XeLaTeX workflow

Contents

1 Overview

2 Commands Reference

3 Fonts & Palettes

4 Boxes & Listings

5 Theorem-like Environments

6 TikZ Tutorial

7 Acknowledgements

What Is Included

This folder contains a reusable Beamer template extracted from the current presentation, with the visual rules moved into a standalone style package.

默认中文字体现已改为完整覆盖的 Songti SC，以保证所有常用简体中文字符都能正确编译；如需楷体，请使用 `\textkai{中文}`。

Core files

- `oucsmartbeamer.sty`: theme, colors, title page, footer, boxes
- `template-manual.tex`: this guide document
- `fonts/`: bundled template fonts
- `badge/`: bundled OUC and Adelaide badges

Use the style file for formatting, and keep your own deck focused on content only.

Folder Layout

```
beamertemplate/
|-- oucsmartbeamer.sty
|-- template-manual.tex
|-- fonts/
|-- badge/
|    |-- AUlightbadge.png
|    |-- ouc.png

```

The default asset paths are now local to the template folder. If you rename or replace the bundled badges, update the corresponding path in `\OUCBeamerSetup`.

Available Commands (1/2)

Command	Purpose
`\OUCTitleFrame`	Draws the custom title slide
`\OUCClosingFrame`	Draws the custom closing slide
`\OUCBeamerSetup{...}`	Adjusts marker text, logos, footer text, closing text
`\OUCSetTitleFontSize{30}{34}`	Sets the cover title font size and line height
`\OUCSetBodyFontSize{\small}`	Sets body text size for normal content boxes and slide text
`\OUCSetHeaderFontSize{\Large}`	Sets the frame title size in the page header
`\OUCSetFooterFontSize{\footnotesize}`	Sets the footer text size
`\OUCSetPalette{N}`	Selects palette 0..9 (brand triple + box backgrounds)
`oucpal1 .. oucpal6`	Six numbered colors exposed by the active palette

Available Commands (2/2)

Command	Purpose
`\textkai{中文}`	Uses the bundled Kai style for selected Chinese text
`infobox`, `warnbox`, `resultbox`, `goalbox`	Reusable content boxes
`The`, `Exa`, `Rmk`, `Pro`, `Le`, `Co`, `De`	Book-style theorem-like environments
`Proof`, `Boxed`	Proof environment with QED, rounded grey container
`Intro`, `RightIntro`	Chapter-introduction boxes (left / right title)
`\UnderlineBox`, `\Quote`	Underlined block, formatted quotation
`\Minipage`, `\EvOdd`	Side-by-side layout, wrap-figure helper
`\Eq`, `\EqL`	Inline align wrappers (unlabelled / labelled)

Font Size Controls

```latex
\OUCSetTitleFontSize{30}{34}
\OUCSetBodyFontSize{\normalsize}
\OUCSetHeaderFontSize{\large}
\OUCSetFooterFontSize{\small}

```

The title size command takes two numbers: font size and baseline skip. The other three commands take standard LaTeX size switches such as `\small`, `\normalsize`, or `\Large`.

Recommended starting point: title `30/34`, body `\normalsize`, header `\large`, footer `\small`.

LaTeX Font Size Switches

Sample text	\tiny
Sample text	\scriptsize
Sample text	\footnotesize
Sample text	\small
Sample text	\normalsize
Sample text	\large
Sample text	\Large
Sample text	\LARGE
Sample text	\huge
Sample text	\Huge

```latex
{\tiny Sample text}
{\scriptsize Sample text}
{\footnotesize Sample text}
{\small Sample text}
{\normalsize Sample text}
{\large Sample text}
{\Large Sample text}
{\LARGE Sample text}
{\huge Sample text}
{\Huge Sample text}

```

Palette Selection

Call \OUCSetPalette{N} once in the preamble. Indices 0–9 map to ten predefined palettes; the active name is exposed via \OUCPaletteName (current: OUC Default).

```
\usepackage{oucsmartbeamer}
\OUCSetPalette{0} %% 0 OUC Default, 1 Brunneophobia, 2 Van Dyke,
                  %% 3 Back in Black, 4 Belle of the Ball,
                  %% 5 Pine Tree,    6 Provence Blue,
                  %% 7 Fresco Blue, 8 Monet,   9 Narcissus

```

Each palette defines six colors oucpal1..oucpal6, sorted light → dark. Slot usage: pal1 = lstlisting background, pal2 = Example / Remark / Intro body background, pal4 = Example / Remark / Intro title background, pal5 = primary (*second* darkest; emphasis color for titles, frametitle, footer, theorem titles). pal3 and pal6 are reserved: in 5-color palettes pal3 holds a duplicate, and pal6 stays as the darkest swatch available for ad-hoc use.

Palette 0 — OUC Default

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

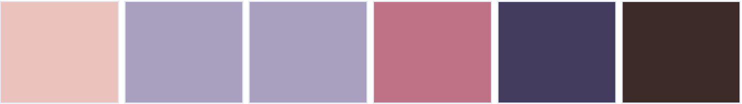
OUC Default — deep navy blue. OUC house palette. `\OUCSetPalette{0}`

Palette 1 — Brunneophobia

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Brunneophobia — warm earth tones, amber and deep brown. `\OUCSetPalette{1}`

Palette 2 — Van Dyke



pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Van Dyke — muted mauve and slate purple with warm rose mid-tones. `\OUCSetPalette{2}`

Palette 3 — Back in Black

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Back in Black — dusty rose fading to near-monochrome charcoal. `\OUCSetPalette{3}`

Palette 4 — Belle of the Ball

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Belle of the Ball — terracotta orange and olive green, bold contrast. `\OUCSetPalette{4}`

Palette 5 — Pine Tree

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Pine Tree — golden amber with muted rose accents and deep forest brown. `\OUCSetPalette{5}`

Palette 6 — Provence Blue

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Provence Blue — sage green and slate blue, subdued Mediterranean tones. `\OUCSetPalette{6}`

Palette 7 — Fresco Blue

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Fresco Blue — bright sky blue grading into deep ocean teal. `\OUCSetPalette{7}`

Palette 8 — Monet

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Monet — soft ivory, dusty coral and forest green, Impressionist palette. `\OUCSetPalette{8}`

Palette 9 — Narcissus

pal1	pal2	pal3	pal4	pal5	pal6
lstlisting	body	accent	title fill	primary	darkest

Narcissus — sandy warm tones with a bold burnt-orange primary. `\OUCSetPalette{9}`

Configuration Keys

```
\OUCBeamerSetup{
    marker=PROJECT REVIEW,
    left logo=badge/AUlightbadge.png,
    right logo=badge/ouc.png,
    footer title=My Deck,
    footer subtitle=Section Name,
    closing title=Questions?,
    closing subtitle=My Deck,
    closing note=Thanks for listening
}

```

Standard Beamer metadata such as `\title`, `\subtitle`, `\author`, `\institute`, and `\date` remain the primary content inputs.

Example Chinese text: 这是一个默认中文示例，可直接用于课题汇报、课程作业和模板说明文档。局部楷体示例：這是楷體文字。

Box Environment Usage

```latex
\begin{infobox}
    Background, definitions, and context.
\end{infobox}

\begin{warnbox}
    Caveats, assumptions, and risks.
\end{warnbox}

\begin{resultbox}
    Findings, conclusions, and key outputs.
\end{resultbox}

\begin{goalbox}
    A short highlighted message.
\end{goalbox}

```

Built-in Environment Preview

Info box
Use for background, definitions, and explanations.

Result box
Use for outcomes, findings, and final numbers.

Warning box
Use for caveats, assumptions, or risks.

Goal boxes work well for short highlighted messages.

Listing Style Usage

```
\begin{frame}[fragile]
    \frametitle{Code Example}
    \begin{lstlisting}[language=Matlab]
for idx = 1:3
    disp(idx)
end
%% the inner lstlisting and the frame are closed here

```

Code slides must use the [fragile] option. The template already applies the bundled `matlabmodern` listing style.

Listing Style Preview

```
for idx = 1:3
    fprintf('Slide %d\n', idx);
end

```

Compile Command

```
cd beamertemplate
latexmk -xelatex -interaction=nonstopmode -halt-on-error template-manual.tex

```

Use XeLaTeX so that the bundled fonts and local badge assets are loaded correctly inside the template folder.

Environment: Definition / Theorem

Definition 1: Group

A set with an associative operation, identity, and inverses.

Theorem 2: Lagrange

If $H \leq G$, then $|H|$ divides $|G|$.

```latex
\begin{De}{Group}{de:g}
  A set with an associative
  operation, identity, inverses.
\end{De}
\begin{The}{Lagrange}{the:lag}
  If $H\le G$, then $|H|$
  divides $|G|$.
\end{The}
```

Environment: Example / Remark

Example 3

$\mathbb{Z}/6\mathbb{Z}$ is cyclic of order 6.

Remark 4

The trivial group $\{e\}$ is the unique group of order 1.

```latex
\begin{Exa}{}{exa:cyc}
  $\mathbb{Z}/6\mathbb{Z}$ is
  cyclic of order $6$.
\end{Exa}
\begin{Rmk}{}{rmk:tr}
  The trivial group $\{e\}$ is
  the unique group of order $1$.
\end{Rmk}
```

Environment: Proposition / Lemma / Corollary

Proposition 5: Closure

$H \cap K$ is closed under the group operation.

Lemma 6: Cancellation

If $ab = ac$ in a group, then $b = c$.

Corollary 7: Order divides

$|\langle g \rangle|$ divides $|G|$.

```
\begin{Pro}{Closure}{pro:cl}
  $H\cap K$ is closed.
\end{Pro}
\begin{Le}{Cancellation}{le:can}
  If $ab=ac$, then $b=c$.
\end{Le}
\begin{Co}{Order divides}{co:ord}
  $|\langle g\rangle|$
  divides $|G|$.
\end{Co}

```

Environment: Proof / Boxed

Proof $x \in H \cap K \Rightarrow x \in H$ and $x \in K$; intersection is a subgroup. $\square$

Rounded grey container suited for paragraph-sized notes or summaries.

```
\begin{Proof}
  $x\in H\cap K \Rightarrow$
  $x\in H$ and $x\in K$.
\end{Proof}
\begin{Boxed}
  Rounded grey container for
  paragraph-sized notes.
\end{Boxed}
```

Environment: Intro / RightIntro

Chapter Intro

Title attached to the upper-left corner; useful as a chapter or section opener.

Right Intro

Mirrored variant with the title on the upper-right corner.

```latex
\begin{Intro}{Chapter Intro}{Title in the
   upper-left corner.}
\end{Intro}
\begin{RightIntro}{Right Intro}
  {Title in the upper-right corner.}
\end{RightIntro}

```

Environment: UnderlineBox / Quote

Underlined block with a double rule.

> *Mathematics is the queen of the sciences.*
>
> C. F. Gauss

```latex
\UnderlineBox{Underlined block
  with a double rule.}
\Quote{Mathematics is the queen
  of the sciences.}{C.\,F.~Gauss}

```

Helpers: Minipage / Eq / EqL

Left.
Side-by-side helper.

Right.
Use widths summing < 1.

$$a^2 + b^2 = c^2$$

$$e^{i\pi} + 1 = 0 \tag{1}$$

```
\Minipage{0.46}{Left content}
         {0.46}{Right content}
\Eq{a^2 + b^2 = c^2}
\EqL{e^{i\pi} + 1 = 0}{eq:euler}
%% \EvOdd{lines}{width}{content}
%%    = right-aligned wrapfigure

```

Environment: infobox / warnbox

Background, definitions, and context.

Caveats, assumptions, and risks.

```
\begin{infobox}
  Background, definitions,
  and context.
\end{infobox}
\begin{warnbox}
  Caveats, assumptions, risks.
\end{warnbox}
```

Environment: resultbox / goalbox

Findings, conclusions, and key outputs.

A short highlighted message.

```latex
\begin{resultbox}
  Findings, conclusions,
  and key outputs.
\end{resultbox}
\begin{goalbox}
  A short highlighted message.
\end{goalbox}
```

Inline Math and Additional Packages

mathtools / amsmath

$\tfrac{N}{D}$ $\tbinom{N}{D}$ \tfrac, \tbinom

$\xrightarrow[below]{above}$ \xrightarrow[b]{a}

$\overbracket{a+b}^{c}$ \overbracket

mathrsfs

$\mathscr{A}$, $\mathscr{B}$, $\mathscr{F}$, $\mathscr{L}$ \mathscr

dsfont

$\mathds{R}$, $\mathds{N}$, $\mathds{Z}$, $\mathds{Q}$ \mathds

polynomial

$1 - x^2$ \polynomial{…}

nicefrac

$\nicefrac{3}{4}$, $\nicefrac{a}{b}$ \nicefrac{}{}

accents

$\underaccent{\leftarrow}{x}$ \underaccent

$\accentset{\star}{y}$ \accentset

dirtytalk

“Hello world” \say{…}

extpfeil

$A \xtwoheadrightarrow{f} B$ \xtwoheadrightarrow

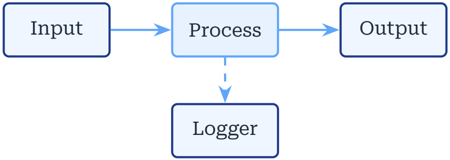
$A \xhookrightarrow{g} B$ \xhookrightarrow

TikZ: Palette-Aware Drawing


```latex
\begin{tikzpicture}[scale=0.75]
  \fill[paleblue]
    (0,0) rectangle (4,1.4);
  \draw[primary,thick,rounded corners=2pt]
    (0,0) rectangle (4,1.4);
  \node[fill=primary,text=white,
        rounded corners=2pt,inner sep=3pt]
    at (2,0.7) {Palette aware};
  \draw[accent,very thick,-Stealth]
    (0,-0.6) -- (4,-0.6);
  \fill[oucpal5] (1.4,-1.8) circle (0.25);
\end{tikzpicture}
```

Tip: use palette-aware names `primary`, `accent`, `paleblue`, `oucpal1–oucpal6` for auto-recoloring. Pre-loaded: `positioning`, `arrows.meta`, `fit`, `calc`, `decorations.pathmorphing`, `shadows.blur`.

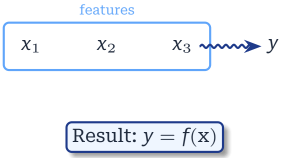
TikZ: Node Diagrams & Arrows




```
\begin{tikzpicture}[
  node distance=6mm and 8mm,

    box/.style={draw=primary,fill=paleblue,
    thick,rounded corners=2pt,
    minimum width=14mm,align=center},
  hi/.style
    ={draw=accent,fill=accent!15,
    thick,rounded corners=2pt,
    minimum width=14mm,align=center},
  arr/.style={-Stealth,accent,thick}]
  \node[box] (a) {Input};
  \node[hi,  right=of a] (b)
    {Process};
  \node[box, right=of b] (c) {Output};
  \node[box, below=of b] (d) {Logger};
  \draw[arr] (a)--(b); \draw[arr]
    (b)--(c);
  \draw[arr,dashed] (b)--(d);
\end{tikzpicture}
```

TikZ: Highlights & Annotations




```latex
\begin{tikzpicture}
  \node (n1) at (0,0)   {$x_1$};
  \node (n2) at (1.0,0) {$x_2$};
  \node (n3) at (2.0,0) {$x_3$};
  \node (n4) at (3.2,0) {$y$};
  \node[draw=accent,thick,
    rounded corners=2pt,
    fit=(n1)(n2)(n3),inner sep=4pt,
    label=above:features] {};

    \draw[primary,thick,-Stealth,decorate,
    decoration={snake,amplitude=.6pt,
    segment length=3pt}]
    (n3.east) -- (n4.west);
  \node[draw=primary,fill=paleblue,
    rounded corners=2pt,
    blur shadow] at (1.5,-1.2)
    {Result: $y=f(\mathbf{x})$};
\end{tikzpicture}
```

Acknowledgements

This template builds on open-source work:

A Tufte-Style Book Template by Kevin Godby *et al.*

`https://www.overleaf.com/latex/templates/a-tufte-style-book-with-vdqi-title-and-contents-page/xqkjxvmrhmmp`

The theorem environments (`The`, `De`, `Exa`, `Rmk`, ...), equation helpers (`\Eq`, `\EqL`), and the mathematical package selection in this template are inspired by and derived from that work.

We gratefully acknowledge Kevin Godby's contribution to the LaTeX community.

Template Ready

OUC Smart Beamer

Use template-manual.tex as your starting point

Haide College, Ocean University of China

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